NATURAL LANGUAGE PROCESSING – ASSIGNMENT 4

Title: Semantic Systems Engineering: Analyse, Represent, Test, Disambiguate and Retrieve
Course Outcome: CO4 – Analyse Semantic Structure and Design Meaning-Based NLP Systems
Total Marks: 50
Nature: Individual Assignment

Student Details

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The answers below follow the assignment's required workflow:

Design → Represent → Implement/Trace → Test → Break → Evaluate → Defend

The assignment does not require source code. The answers use semantic frames, predicate logic, parse alternatives, sense inventories, retrieval scores and evaluation calculations.

TASK 1 – REPRESENTING MEANING AND LANGUAGE STRUCTURE (8 Marks)

Sentence: “Leena sent the report to Arun after the meeting.”

1.1 Semantic Structure Analysis (2 Marks)

Meaning components

Component	Identification	Explanation
Agent	Leena	Person performing the action
Action	Sent	Main event or activity
Theme	The report	Object being transferred
Recipient	Arun	Person receiving the report
Temporal expression	After the meeting	Indicates when the sending occurred
Event	Sending event	Complete action performed by Leena

Semantic roles

Agent: Leena

Theme: Report

Recipient: Arun

Time: After the meeting

Surface form and meaning

The surface form is the actual arrangement of words in the sentence.

The semantic representation describes the underlying meaning and relationships between the participants.

For example:

Surface form: Leena sent the report to Arun after the meeting.

Meaning: Leena performed a sending event, the report was transferred, Arun was the recipient, and the event occurred after the meeting.

The sentence does not specify the exact date, location, delivery method or whether Arun successfully opened the report.

1.2 Build Meaning Representations (3 Marks)

A. Semantic-role/event frame

EVENT: SEND-01

Slot	Value
Event type	Sending
Agent	Leena
Theme	Report
Recipient	Arun
Time	After the meeting
Tense	Past

This frame clearly identifies the event and its participants.

B. Graph representation

Semantic relationship graph

SEND EVENT

Agent

Leena

Theme

Report

Recipient

Arun

Time: After the meeting

The graph uses nodes for entities and events and labelled edges for semantic relationships.

Comparison

Representation	Information captured
Event frame	Event type, agent, theme, recipient and time
Semantic graph	Entities, event node and relationships between them

The event frame is easy to store in a database. The graph is useful for connecting events and entities across multiple sentences.

1.3 Meaning-Preserving Variation (2 Marks)

Original sentence:

“Leena sent the report to Arun after the meeting.”

Alternative sentence:

“After the meeting, the report was sent to Arun by Leena.”

The sentence structure changes from active voice to passive voice, but the core semantic roles remain the same.

Semantic role	Original	Alternative
Agent	Leena	Leena
Theme	Report	Report
Recipient	Arun	Arun
Time	After the meeting	After the meeting

The following changes would alter the meaning:

> “Leena sent the report to Ravi” changes the recipient.
>
> “Arun sent the report to Leena” changes the agent and recipient.
>
> “Leena sent the report before the meeting” changes the time relation.

1.4 Representation Check (1 Mark)

A validation rule can check that every sending event has an agent and a theme.

```
IF event_type = SEND:
   REQUIRE agent
   REQUIRE theme
   REQUIRE recipient
   REQUIRE time
```

If the recipient is missing, the system should flag the event as incomplete rather than silently assuming a recipient.

Result: The event frame and semantic graph preserve the important meaning of the sentence independently of its surface word order.

TASK 2 – FIRST ORDER PREDICATE CALCULUS: FORMALISE AND TEST (9 Marks)

2.1 Predicate Vocabulary (2 Marks)

First Order Predicate Calculus (FOPC) is a formal language used to represent objects, properties and relationships using predicates, variables, constants and quantifiers.

Constants

Constant	Meaning
NLP	Natural Language Processing course
C1	A course offered by the department
C2	Another department course

Variables

\(x,y\): Learners

\(c,d\): Courses

Predicates

Predicate	Meaning	Argument order
Learner(x)	x is a learner	Learner
Course(c)	c is a course	Course
Enrolled(x,c)	x is enrolled in c	Learner, Course
Completed(x,c)	x completed c	Learner, Course
Prerequisite(c,d)	d is a prerequisite of c	Course, Prerequisite course

Additional predicate

\[OfferedBy(c,Dept) \]

Meaning: Course \(c\) is offered by the department.

2.2 Formalisation (3 Marks)

Statement 1

“Every learner enrolled in NLP has completed at least one prerequisite.”

Formula:

\[\forall x[(Learner(x)\land Enrolled(x,NLP)) \rightarrow \exists c(Course(c)\land Prerequisite(NLP,c)\land Completed(x,c))] \]

Explanation:

For every learner, if that learner is enrolled in NLP, there must be at least one course that is a prerequisite of NLP and that the learner has completed.

Statement 2

“Some learner completed every course offered by the department.”

Formula:

$$\exists x[Learner(x)\land \forall c((Course(c)\land OfferedBy(c,Dept)) \rightarrow Completed(x,c))]$$

Explanation:

There is at least one learner who has completed every course offered by the department.

Statement 3

“No learner completed a course that they were not enrolled in.”

Formula:

$$\forall x\forall c[(Learner(x)\land Course(c)\land Completed(x,c)) \rightarrow Enrolled(x,c)]$$

Equivalent form:

$$\neg\exists x\exists c[Learner(x)\land Course(c)\land Completed(x,c)\land\neg Enrolled(x,c)]$$

Explanation:

There is no learner-course pair where the learner completed a course without being enrolled in it.

2.3 Quantifier-Scope Challenge (2 Marks)

Consider:

$$\forall x\exists c\ Enrolled(x,c)$$

Meaning: Every learner is enrolled in at least one course.

Compare with:

$$\exists c\forall x\ Enrolled(x,c)$$

Meaning: There is one particular course in which every learner is enrolled.

These formulas are not logically equivalent.

Counterexample model

Assume there are two learners and two courses:

Learner	Course enrolled
A	NLP
B	AI

Formula 1 is true because every learner has at least one course.

Formula 2 is false because there is no single course shared by both learners.

Therefore, changing the order of quantifiers changes the meaning.

2.4 Consistency Test (2 Marks)

Knowledge base

Facts:

Learner(Asha)
Learner(Ravi)
Course(NLP)
Course(AI)
Enrolled(Asha,NLP)
Enrolled(Ravi,AI)
Completed(Asha,NLP)

Rule:

\[\forall x\forall c[Completed(x,c)\rightarrow Enrolled(x,c)] \]

Query

Has Asha completed NLP?

\[Completed(Asha,NLP) \]

The knowledge base contains this fact directly.

Query follows

Inference trace

Read the fact Completed(Asha,NLP).

Match it with the query.

The query is directly supported by the knowledge base.

Return TRUE.

Additional query:

Has Ravi completed NLP?

\[Completed(Ravi,NLP) \]

There is no supporting fact or rule that establishes this statement.

Therefore, it is not derivable from the knowledge base. This does not automatically prove it is false.

Result: FOPC provides a precise way to represent learner-course relationships and test logical conclusions.

TASK 3 – LINGUISTIC CONCEPTS AND SYNTAX-DRIVEN SEMANTIC ANALYSIS (9 Marks)

Sentence: “The analyst noticed the student with the camera near the library.”

3.1 Linguistically Relevant Concepts (2 Marks)

Concept identification

Concept	Example	Meaning
Entity 1	Analyst	Person observing or noticing
Entity 2	Student	Person noticed
Action	Noticed	Main event
Instrument	Camera	Object used for observation or recording
Location	Library	Place near which the event or student is situated
Modifier	The	Identifies a particular entity
Preposition	With	Can indicate instrument or possession/accompaniment
Preposition	Near	Indicates spatial relationship

Semantic roles

Agent/Experiencer: Analyst.

Theme/Perceived entity: Student.

Instrument: Camera, if attached to the noticing event.

Location: Library, depending on attachment.

These concepts matter because the system must identify who performed the action, who was noticed, what object was involved and where the event or person was located.

3.2 Parse-to-Meaning Procedure (3 Marks)

The phrase “with the camera near the library” has more than one possible syntactic attachment.

Interpretation 1: Camera is attached to the student

Meaning: The analyst noticed the student who had a camera. The student was near the library.

Bracketed structure:

```
[S
  [NP The analyst]
  [VP noticed
    [NP
      [NP the student]
      [PP with the camera]
      [PP near the library]]]]
```

Semantic representation:

```
NOTICE(
    Agent = Analyst,
    Theme = Student,
    StudentPossesses = Camera,
    StudentLocation = Near(Library)
)
```

Interpretation 2: Camera is attached to the noticing event

Meaning: The analyst used a camera to notice or observe the student near the library.

Bracketed structure:

```
[S
  [NP The analyst]
  [VP
    [VP noticed
      [NP the student]]
    [PP with the camera]
    [PP near the library]]]
```

Semantic representation:

```
NOTICE(
    Agent = Analyst,
    Theme = Student,
    Instrument = Camera,
    Location = Near(Library)
)
```

The second interpretation assumes the camera is an instrument and the location applies to the noticing event.

Syntax-driven procedure

Tokenise the sentence into individual words.

Identify the noun phrase the analyst.

Identify the verb noticed.

Identify the object noun phrase the student.

Identify the PP with the camera.

Generate possible PP attachment positions.

Identify the PP near the library.

Construct a semantic representation for each parse.

Use context and selectional preferences to rank the interpretations.

The interpretations diverge at the attachment of with the camera: it modifies the student in Interpretation 1 and the noticing event in Interpretation 2.

3.3 Composition and Type/Role Checks (2 Marks)

Step-by-step semantic composition

Step	Input phrase	Semantic result
1	The analyst	Entity: Analyst
2	The student	Entity: Student
3	Noticed	Event: NOTICE
4	With the camera	Instrument or possession relation
5	Near the library	Spatial relation
6	Complete sentence	Combined event representation

Semantic composition

The verb noticed creates an event with an experiencer and a perceived object.

\[NOTICE(Analyst,Student) \]

For Interpretation 1:

\[Has(Student,Camera) \land Near(Student,Library) \]

For Interpretation 2:

\[Uses(Analyst,Camera) \land Near(NOTICE_EVENT,Library) \]

Two validation checks

Check 1 – Semantic role compatibility

A camera can be an instrument used for observation. The system should verify that the instrument is connected to a suitable event or entity.

Check 2 – Attachment compatibility

The system should check whether the PP modifies a noun phrase or verb phrase according to the selected parse. It should not assign the same PP to two incompatible attachment positions in one interpretation.

3.4 Error Challenge (2 Marks)

Example of an implausible interpretation

Sentence:

“The analyst noticed the student with the building.”

A parser may attach with the building to the student and produce:

Has(Student,Building)

This is syntactically possible in some contexts but may be semantically implausible.

Possible errors

Incorrect instrument assignment: The system may treat the camera as belonging to the analyst when the sentence means that the student has the camera.

Incorrect location assignment: The system may attach near the library to the wrong noun phrase.

Refinement

Use selectional preferences and context.

For example:

use(camera, observe) is a plausible event-instrument relationship.

near(student, library) is a plausible spatial relationship.

If the sentence remains ambiguous, retain both interpretations and request clarification.

Result: Syntax-driven semantic analysis generates alternative meanings and uses semantic compatibility to reduce incorrect interpretations.

TASK 4 – LEXEMES, SENSES AND WORD SENSE DISAMBIGUATION (8 Marks)

The target lexeme is LIGHT.

4.1 Lexeme and Sense Inventory (2 Marks)

A lexeme is an abstract vocabulary unit that groups related word forms and meanings. A word form is the actual written or spoken form used in a sentence. A sense is one particular meaning of a lexeme.

Sense inventory

Sentence	Word form	Part of speech	Sense
Turn on the light.	light	Noun	Lamp or source of illumination
This bag is light.	light	Adjective	Having low weight
The light meal was served early.	light	Adjective	Small or not heavy in quantity

Explanation

Sentence A: “Turn on the light.”

Here, light is a noun referring to an electrical lighting device or source of illumination.

Sentence B: “This bag is light.”

Here, light is an adjective describing the bag's low weight.

Sentence C: “The light meal was served early.”

Here, light is an adjective meaning a meal that is not heavy or substantial.

The same written form can represent different grammatical categories and meanings.

4.2 WSD Design (2 Marks)

Proposed method: Rule-based contextual WSD

WSD means Word Sense Disambiguation. It identifies the intended meaning of a word based on its surrounding context.

Input: A sentence containing the word light.

Output: The selected sense label.

Candidate senses

Sense ID	Meaning
L1	Lighting device or illumination
L2	Low weight
L3	Small or not heavy meal/food

Context features

The system checks:

Part of speech.

Nearby verbs.

Nearby nouns.

Adjectives and modifiers.

Common word combinations.

Decision rules

```
IF light follows "turn on", "switch on", or "lamp":
   Select L1

ELSE IF light describes bag, box, object, or weight:
   Select L2

ELSE IF light modifies meal, food, snack, or breakfast:
   Select L3

ELSE:
   Return UNKNOWN or ask for clarification
```

Output format

```
Input: This bag is light.
Word: light
POS: Adjective
Selected sense: L2
Meaning: Low weight
Confidence: High
```

The confidence value is a qualitative label in this manual design, not a measured probability.

4.3 Apply and Challenge (2 Marks)

Test sentence A

“Turn on the light.”

Evidence:

The phrase turn on commonly occurs with a lighting device.

The word the introduces a noun phrase.

Selected sense: L1 – Lighting device.

Test sentence B

“This bag is light.”

Evidence:

Bag is an object.

Light describes its weight.

Selected sense: L2 – Low weight.

Test sentence C

“The light meal was served early.”

Evidence:

Meal is a food-related noun.

Light describes the amount or heaviness of the meal.

Selected sense: L3 – Not heavy or substantial.

Original sentence D

"The room became bright when we switched on the light."

Evidence:

Switched on is associated with illumination.

Room provides a lighting context.

Selected sense: L1 – Source of illumination.

Failure example

"The light was difficult to carry."

This sentence may refer to a lighting device that is difficult to carry, or it may contain an unusual use of light as a noun referring to something lightweight.

The rule-based method may incorrectly choose L2 because it sees carry.

Improvement: Use a larger context window and syntactic analysis to identify whether light is a noun or adjective.

4.4 Evaluation (2 Marks)

Eight-instance labelled test set

No.	Sentence	Expected sense
1	Turn on the light.	L1
2	Switch off the light.	L1
3	The light is shining.	L1
4	This bag is light.	L2
5	The box is light.	L2
6	This laptop is light.	L2
7	We had a light meal.	L3
8	A light breakfast is enough.	L3

Assume the rule-based method correctly classifies all eight examples in this designed test.

\[Accuracy=\frac{Correct\ Predictions}{Total\ Predictions}\times100 \]
\[Accuracy=\frac{8}{8}\times100=100\% \]
Illustrative test accuracy: 100%

This is a manually constructed test result, not evidence of performance on a real-world dataset.

Why high accuracy may be misleading

The test set is very small.

The examples are simple and clearly labelled.

Real sentences may contain unfamiliar contexts.

A balanced test set may not reflect real-world frequency.

Ambiguous examples may have more than one acceptable interpretation.

For reliable evaluation, use a larger dataset with diverse sentences and independent human annotations.

Result: The rule-based WSD method distinguishes the three senses of light using grammatical and contextual evidence.

TASK 5 – SEMANTIC INFORMATION RETRIEVAL (8 Marks)

Query: “How can a system identify the intended meaning of a word in context?”

5.1 Query and Document Semantics (2 Marks)

Query interpretation

The user's intention is to find information about identifying the correct meaning of an ambiguous word based on the sentence in which it appears.

Central concepts

Concept	Meaning
Word meaning	Meaning of a vocabulary item
Context	Surrounding words and sentence
Ambiguity	Multiple possible meanings
Word Sense Disambiguation	Selecting the intended sense
NLP system	Computer system processing language

Query representation

QUERY:
Main topic = Word Sense Disambiguation
Purpose = Identify intended word meaning
Context = Surrounding sentence
Method = Context-based semantic analysis

Candidate documents

Document ID	Document title/content	Relevance
D1	Introduction to Word Sense Disambiguation	Relevant
D2	Contextual methods for identifying word meanings	Relevant
D3	Dictionary-based semantic analysis	Relevant
D4	How to improve computer network speed	Not relevant
D5	Ambiguous words and their linguistic meanings	Relevant
D6	General introduction to Python programming	Not relevant

These documents form a small illustrative retrieval collection.

5.2 Ranking Strategy (2 Marks)

The search engine uses two signals:

Lexical similarity – checks matching words.

Semantic similarity – checks whether the document expresses the same meaning as the query.

Ranking formula

$$ Score(D,Q)=0.4L+0.6S $$

Where:

L = Lexical similarity.

S = Semantic similarity.

0.4 = Lexical weight.

0.6 = Semantic weight.

Semantic similarity receives greater weight because the task is meaning-based retrieval.

5.3 Retrieval Trace (2 Marks)

Example similarity scores

The following values are illustrative scores assigned to demonstrate the ranking method.

Document	Lexical score (L)	Semantic score (S)	Final score
D1	0.90	0.95	0.93
D2	0.55	0.98	0.81
D3	0.65	0.80	0.74
D4	0.20	0.10	0.14
D5	0.70	0.90	0.82
D6	0.15	0.20	0.18

Calculation example

For D1:

$$ Score=0.4(0.90)+0.6(0.95) $$
$$ =0.36+0.57 $$
$$ =0.93 $$

For D2:

$$ Score=0.4(0.55)+0.6(0.98) $$
$$ =0.22+0.588 $$
$$ =0.808\approx0.81 $$

Final ranking

Rank	Document	Score
1	D1 – Introduction to Word Sense Disambiguation	0.93
2	D5 – Ambiguous Words and Their Linguistic Meanings	0.82
3	D2 – Contextual Methods for Identifying Word Meanings	0.81
4	D3 – Dictionary-Based Semantic Analysis	0.74
5	D6 – General Introduction to Python Programming	0.18
6	D4 – How to Improve Computer Network Speed	0.14

D2 has lower literal word overlap than D1, but its semantic similarity is very high. Therefore, it receives a strong ranking.

5.4 Evaluate (2 Marks)

Precision

Precision measures how many retrieved documents are relevant.

$$ Precision=\frac{Relevant\ Retrieved}{Total\ Retrieved} $$

Recall

Recall measures how many relevant documents were retrieved from all relevant documents.

$$ Recall=\frac{Relevant\ Retrieved}{Total\ Relevant} $$

Assume the top four documents are retrieved.

Retrieved documents: D1, D5, D2, D3.

Relevant documents: D1, D2, D3, D5.

Total relevant documents: 4.

Relevant retrieved documents: 4.

Precision:

\[Precision=\frac{4}{4}=1=100\% \]

Recall:

\[Recall=\frac{4}{4}=1=100\% \]

False positive and false negative

A false positive is a document returned by the system that is not relevant.

Example: A document about general dictionary definitions may discuss word meanings but not explain how to identify the intended sense in context.

A false negative is a relevant document that the system fails to retrieve.

Example: A document titled “Contextual Interpretation of Ambiguous Vocabulary” may be relevant even though it does not use the exact phrase word sense disambiguation.

Improvement

Use contextual embeddings, concept matching and sense compatibility. The system should also check whether the document answers the user's actual question instead of relying only on shared vocabulary.

Result: Semantic retrieval improves document discovery by considering both lexical matching and contextual meaning.

TASK 6 – INTEGRATED SEMANTIC ANALYSIS DESIGN CHALLENGE (8 Marks)

Sentence: “I watched the crane by the bank, then it moved.”

The system must understand the intended meanings of crane and bank, identify the reference of it and retrieve a suitable information article.

6.1 Model the Ambiguity (1 Mark)

Possible meanings of crane

Sense	Meaning
Crane 1	A large construction machine
Crane 2	A large bird

Possible meanings of bank

Sense	Meaning
Bank 1	Financial institution
Bank 2	Land beside a river

Pronoun ambiguity

The word it may refer to:

The crane machine.

The bird.

The system needs additional information about the scene, nearby objects and the intended topic.

6.2 Candidate Meaning Structures (2 Marks)

Interpretation 1: Construction crane near a financial bank

Meaning:

A person watched a construction crane located near a financial institution, and the crane machine moved.

Semantic roles:

Observer: I.

Theme: Crane machine.

Location: Near financial bank.

Action: Watched.

Second action: Moved.

Pronoun reference: Crane machine.

Structured representation:

```
EVENT 1:
   Type = WATCH
   Observer = User
   Object = Crane_Machine
   Location = Near(Financial_Bank)

EVENT 2:
   Type = MOVE
   Entity = Crane_Machine
   Time = After(EVENT 1)
```

Interpretation 2: Bird near a river bank

Meaning:

A person watched a crane bird standing near the bank of a river, and the bird moved.

Semantic roles:

Observer: I.

Theme: Crane bird.

Location: River bank.

Action: Watched.

Second action: Moved.

Pronoun reference: Crane bird.

Structured representation:

```
EVENT 1:
  Type = WATCH
  Observer = User
  Object = Crane_Bird
  Location = Near(River_Bank)

EVENT 2:
  Type = MOVE
  Entity = Crane_Bird
  Time = After(EVENT 1)
```

Comparison

Feature	Interpretation 1	Interpretation 2
Crane sense	Construction machine	Bird
Bank sense	Financial institution	River bank
Location	Near a bank building	Near a river
Moved	Machine moved	Bird moved
Pronoun it	Machine	Bird

6.3 Syntax and WSD Trace (2 Marks)

Step-by-step interpretation

Tokenise the sentence: I / watched / the / crane / by / the / bank / then / it / moved.

Identify watched as the main verb.

Identify I as the observer.

Identify the crane as the object of watched.

Identify by the bank as a prepositional phrase describing the location of the crane or the watching event.

Identify it as a pronoun referring to an earlier entity.

Identify moved as the second action.

Generate the two candidate interpretations

Use context and sense compatibility to rank them.

Contextual evidence

Evidence	Construction interpretation	Bird interpretation
Building site	Strong support	Weak support
River	Weak support	Strong support
Construction workers	Strong support	Weak support
Wetland habitat	Weak support	Strong support
Crane operator	Strong support	Weak support

Pronoun resolution

The system should check the grammatical number and semantic compatibility of possible antecedents.

Both the crane machine and the crane bird are singular and can move. Therefore, grammar alone cannot resolve the pronoun.

If evidence is insufficient, the assistant should preserve both interpretations or ask:

“Do you mean a construction crane or a crane bird?”

6.4 Semantic Retrieval (2 Marks)

After selecting the intended interpretation, the system converts the meaning into a semantic search query.

Example: Selected construction interpretation

QUERY:
Topic = Construction crane
Action = Movement
Context = Crane operating near a building
Intent = Information about crane movement and operation

Candidate documents

Document	Content	Relevance
D1	Construction Crane Movement and Safety	Relevant
D2	How Tower Cranes Operate	Relevant
D3	Crane Birds in River Habitats	Wrong crane sense
D4	Banking Services and Account Operations	Wrong bank sense
D5	Crane Inspection and Mechanical Movement	Relevant

Ranking formula

$$ Score=0.4L+0.6S $$

Where:

L = Lexical similarity.

S = Semantic similarity.

Illustrative retrieval scores

Document	Lexical score	Semantic score	Final score
D1	0.85	0.95	0.91
D2	0.70	0.92	0.83
D3	0.90	0.20	0.48
D4	0.60	0.05	0.27
D5	0.75	0.90	0.84

Example calculation for D1:

$$ Score=0.4(0.85)+0.6(0.95) $$
$$ =0.34+0.57=0.91 $$

Final ranking

Rank	Document	Final score
1	D1 – Construction Crane Movement and Safety	0.91
2	D5 – Crane Inspection and Mechanical Movement	0.84
3	D2 – How Tower Cranes Operate	0.83
4	D3 – Crane Birds in River Habitats	0.48
5	D4 – Banking Services and Account Operations	0.27

D3 contains the word crane, but it refers to the bird. The semantic score is low because the selected interpretation is about construction machinery.

Similarly, D4 contains the word bank, but it is not relevant to the selected construction interpretation.

6.5 Adversarial Refinement (1 Mark)

Misleading context

Suppose the previous conversation was about wetland birds, but the sentence is:

"I watched the crane by the bank, then it moved."

The system may incorrectly select the bird interpretation because it relies too heavily on previous conversation context.

Predicted system response

The system may rank the bird-related document higher even if the user intended a construction machine.

Refinement

Use multiple contextual signals:

Current sentence context.

Previous conversation context.

Semantic compatibility.

Entity type and location relationships.

Confidence threshold.

OVERALL CONCLUSION

This assignment demonstrates how semantic analysis converts natural-language sentences into structured computer-readable representations. Semantic roles and event frames identify participants and actions. First Order Predicate Calculus represents relationships and supports logical inference. Syntax-driven semantic analysis explains how sentence structure influences meaning and attachment.